Spotify Popularity Prediction Using a Two-Stage Machine Learning Pipeline

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Abstract. This study presents a reproducible machine learning pipeline for predicting song popularity on Spotify using the Spotify Tracks Dataset, which contains 114,000 records and 20 descriptive variables. The proposed pipeline includes data ingestion, cleaning, duplicate handling, feature preparation, exploratory analysis, model training, and performance evaluation. Instead of framing the task only as a conventional regression problem, the study addresses the strong concentration of zero-popularity records through a Two-Stage Hurdle approach. First, the model estimates whether a track can achieve a minimum level of relevance by distinguishing songs with zero popularity from those with positive popularity. Second, among songs that overcome this initial barrier, the model evaluates whether a track can reach the top 20% of popularity. The dataset was divided into training and test sets using an 80/20 stratified split to preserve the proportion of zero and non-zero popularity records. Regression baselines showed limited predictive capacity, with XGBoost reaching an R^2 of 0.1526. In contrast, the first stage of the Hurdle strategy achieved strong performance using Random Forest, obtaining an F_1-score of 0.9631. However, predicting high-impact hits remained challenging, with the combined Hurdle model reaching a ROC-AUC of 0.6865 and an Average Precision of 0.3352. These findings suggest that acoustic features are useful for identifying minimum relevance, but they are insufficient to fully explain commercial success without contextual, temporal, and marketing-related variables.

Keywords: Spotify, popularity, prediction, machine learning, hurdle model, zero-inflation, audio features, reproducibility

1 Introduction

The music industry has experienced a significant transformation driven by the growth of streaming platforms. Spotify, with a user base exceeding 600 million active users, has become a leading digital ecosystem for the distribution and consumption of music content. In this context, quantitative analysis has become increasingly relevant for supporting decision-making in the music industry. The ability to identify patterns associated with market performance can help artists, producers, and record labels reduce uncertainty and better understand the factors related to song popularity.

Historically, decision-making in record labels and among independent artists has relied on experience-based judgment or on reactive monitoring of trends. However, the expansion of digital music platforms has generated large volumes of structured information that can be analyzed computationally. Spotify quantifies technical attributes for each track, generating datasets that can be used to study the relationship between musical characteristics and popularity. The central problem lies in the difficulty of processing these records in order to extract actionable knowledge. Therefore, this study addresses the following research question: is it possible to predict the popularity of a song on Spotify based on measurable audio features such as energy, danceability, tempo, valence, and acousticness? Answering this question requires a reproducible data processing workflow and an experimental design that allows the evaluated models to be compared consistently.

2 Related Work

Music popularity prediction has been studied using different machine learning approaches, particularly with datasets derived from Spotify audio features [9]. Khan et al. [5] analyzed the effect of feature selection on music popularity classification using Spotify data and machine learning algorithms. Their study compared models built with all features against models using selected features and showed that feature selection can reduce computational complexity while maintaining classification performance. This finding is relevant for music analytics because it suggests that not all available audio descriptors contribute equally to popularity prediction, and that selecting informative musical variables can improve the interpretability and efficiency of predictive models.

Saragih [8] examined song popularity prediction using Spotify audio features from Indonesian streaming users. The study compared both regression and classification algorithms and reported that Extra Trees Regressor and Random Forest Classifier achieved the best performance in their respective tasks. This work is closely related to the present study because it evaluates the predictive capacity of Spotify acoustic descriptors and highlights Random Forest as an effective classification model for popularity-related tasks. However, it also shows that the effectiveness of audio features may depend on market context and that popularity prediction remains a complex problem influenced by cultural and consumption patterns.

Other studies have emphasized that audio features alone may not fully explain whether a song becomes a hit [6]. Zhao et al. [11] investigated hit song prediction by combining Spotify audio features with lyrics and engineered metadata features, such as popularity continuity, genre class, and song title topic. Their results showed that Random Forest and Logistic Regression achieved stronger performance when multiple feature sources were combined. This supports the idea that contextual and metadata-based variables can improve prediction beyond the information captured by acoustic descriptors alone. Therefore, models

based only on static musical features may have limited capacity to identify high-impact hits.

In addition to the challenges associated with music popularity prediction, the target variable in this study presents a strong concentration of zero values and an imbalanced class distribution [2]. This type of distribution is related to the broader problem of zero-inflated data, where standard predictive models may be affected by the high frequency of zero outcomes. Ben Khedher and Yun [1] proposed an interpretable machine learning-based Hurdle model for zero-inflated crash frequency data, using a two-stage strategy to first model the occurrence of a non-zero event and then estimate the positive outcome. Although their application domain differs from music analytics, their methodological approach supports the use of two-stage modeling when zero and non-zero outcomes may be generated by different processes.

Based on these previous works, the present study contributes to the literature by combining music popularity prediction with a two-stage machine learning strategy designed to address zero-inflated target behavior. Unlike studies that treat Spotify popularity mainly as a direct regression or standard classification problem, this work first distinguishes tracks with zero popularity from those with positive popularity and then evaluates whether relevant tracks can reach the top 20% of popularity. This approach provides a more interpretable analytical framework for datasets where a substantial portion of records remains at zero popularity.

3 Materials and Methods

This section describes the methodological framework, dataset, data preparation process, experimental design, and technological environment used to implement the proposed two-stage machine learning pipeline. The objective was to ensure that the workflow could be reproduced and that the evaluated models were compared under consistent experimental conditions.

3.1 Methodological Framework

This study follows the CRISP-DM methodology, which stands for Cross-Industry Standard Process for Data Mining. According to IBM [4], this framework organizes data mining projects into structured phases, including business understanding, data understanding, data preparation, modeling, evaluation, and deployment. In this work, CRISP-DM was used to guide the process from dataset exploration and preprocessing to model training and performance evaluation.

3.2 Dataset Description

This project uses the Spotify Tracks Dataset published by Pandya [7]. The dataset contains metadata and audio-related variables that describe musical characteristics of individual tracks. It is characterized by the following elements:

- **Volume:** 114,000 records.
- **Dimensions:** 20 columns, including metadata, acoustic descriptors, and the popularity target variable.
- **Study variables:** Continuous musical attributes such as *energy*, *danceability*, *tempo*, *valence*, and *acousticness*, among others.
- **Target variable:** *popularity*, which represents the popularity score assigned to each track in the dataset.

The practical motivation of this study is to provide a data-driven approach that helps relate measurable audio attributes to potential market performance on Spotify. From an academic perspective, the study evaluates the usefulness and limitations of machine learning models when the target variable presents a strong concentration of zero values.

3.3 Data Pipeline Design

The proposed pipeline was designed as a reproducible workflow for processing, cleaning, transforming, modeling, and evaluating Spotify track data. As shown in Fig. 1, the process follows a sequential structure that starts with the original dataset and continues through data cleaning, feature preparation, train-test splitting, regression baseline evaluation, Two-Stage Hurdle modeling, model evaluation, and final result interpretation.

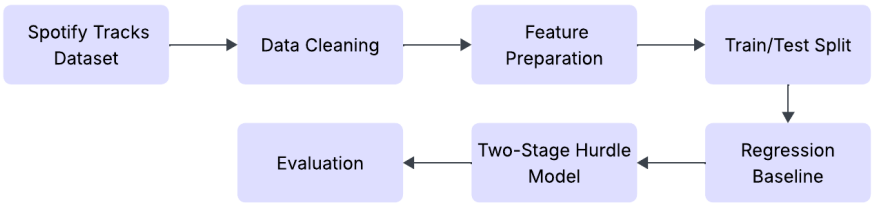


Fig. 1. Workflow of the proposed two-stage machine learning pipeline for Spotify popularity prediction.

The process was organized into the following stages:

- **Data ingestion:** Loading the original Spotify Tracks Dataset into the development environment.
- **Data cleaning:** Removing duplicated records and handling missing values to improve data quality.
- **Feature preparation:** Selecting musical characteristics as input variables and preparing the *popularity* variable as the prediction target.
- **Train-test split:** Dividing the dataset into training and test sets using an 80/20 stratified hold-out strategy.

- **Regression baseline:** Evaluating initial regression models to determine whether acoustic features could directly predict the numerical popularity score.
- **Two-Stage Hurdle modeling:** Reformulating the prediction task into two classification stages to address the concentration of zero-popularity records.
- **Model evaluation:** Comparing model performance using regression, classification, and threshold-based metrics.

This structure was adopted to make the workflow easier to reproduce, inspect, and modify in future iterations.

3.4 Data Preparation

The predictive modeling process was based on the musical characteristics available in the Spotify Tracks Dataset. These variables were selected as input features because they describe measurable acoustic and musical properties of each track. The target variable was *popularity*.

Before training the models, the dataset was cleaned by removing duplicated records and handling missing values. After this process, the original dataset of 114,000 records was reduced to 83,494 unique records. This step was necessary because some tracks shared identical acoustic characteristics across different records, which could have introduced bias or inflated model performance.

Due to the high concentration of songs with a popularity value equal to zero, an auxiliary binary variable was created to support the train-test split. This variable assigned a value of 0 to songs with zero popularity and a value of 1 to songs with popularity greater than zero. The purpose of this auxiliary variable was not to replace the original target, but to preserve the proportion of zero and non-zero popularity records during the data partition.

The complete dataset was divided into training and test sets using an 80/20 hold-out strategy. The split was stratified according to the auxiliary zero-popularity variable, ensuring that both subsets maintained a similar distribution of songs with zero popularity and songs with positive popularity. A fixed random seed was used to make the partition reproducible.

3.5 Model Training and Evaluation

The experimental design included both regression and classification models. First, regression models were trained as baselines to evaluate whether musical characteristics could directly predict the numerical popularity score. The evaluated regression models included Linear Regression, Random Forest, and XGBoost. Their performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2).

After observing the limited explanatory capacity of the regression baseline and the strong concentration of zero-popularity records, the problem was reformulated using a Two-Stage Hurdle strategy. In the first stage, the objective was

to classify whether a song would remain at zero popularity or achieve a minimum level of relevance, defined as a popularity value greater than zero. In the second stage, only songs that passed the initial relevance barrier were analyzed to determine whether they could reach the top 20% of popularity. In this study, the hit threshold corresponded to a popularity score of 54, equivalent to the 80th percentile of the analyzed distribution.

The classification models were trained using the same input features to allow a consistent comparison between algorithms. For the first stage, Logistic Regression, Decision Tree, and Random Forest models were evaluated. For the second stage, Decision Tree, Random Forest, and XGBoost models were compared. Model performance was assessed using Accuracy, Precision, Recall, and F_1-Score. For the combined Hurdle model, ROC-AUC, Average Precision, the optimal classification threshold, and the confusion matrix were also analyzed, considering the limitations of accuracy-based evaluation under imbalanced class distributions [2].

Model evaluation was conducted using the independent test set obtained from the 80/20 split. Cross-validation was not applied in this stage of the project. Therefore, the reported metrics correspond to a hold-out validation strategy. Future work may incorporate cross-validation or repeated stratified validation schemes to further assess the stability and generalization capacity of the proposed models.

3.6 Technological Environment and Reproducibility

GitHub Codespaces was used as a collaborative cloud-based development environment to support the implementation and reproducibility of the proposed pipeline [3]. Codespaces provided a standardized workspace where the project dependencies, source code, notebooks, and configuration files could be managed consistently. This environment also facilitated version control through GitHub repositories, allowing changes in the data processing and modeling workflow to be tracked throughout the development process.

The computational workflow was implemented in Python. The main libraries used included `pandas` and `numpy` for data manipulation and numerical processing; `scikit-learn` for data splitting, preprocessing, model training, and metric evaluation; `xgboost` for gradient boosting models; and `matplotlib` and `seaborn` for exploratory and performance visualizations. Additionally, `kaggle` was used to facilitate dataset access, while `ipykernel` supported notebook-based experimentation within the development environment.

To improve reproducibility, the required dependencies were specified in a `requirements.txt` file. This allowed the experimental environment to be recreated with the same core libraries used during data preparation, model training, and evaluation. In this way, the technological setup supported a reproducible machine learning workflow rather than relying on a traditional large-scale distributed computing infrastructure.

4 Results

In this study, data analysis was structured to examine the relationship between intrinsic audio properties and market performance on Spotify. The following subsections present the findings derived from the implementation of regression and classification models under a reproducible machine learning approach.

4.1 Regression Models

Establishing a baseline through regression models allows a direct quantification of the relationship between musical characteristics and popularity. Based on the results of these models, a different analytical approach was later considered in order to understand the magnitude of the proposed problem. For this analysis, the original dataset of 114,000 records was subjected to a process of cleaning and consolidation of acoustic duplicates, resulting in a final dataset of 83,494 unique records. In other words, some songs had identical musical characteristics but appeared in different records, which could have inflated the final results.

The regression models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2), as shown in Table 1.

Table 1. Performance metrics of regression models.

Model	RMSE	MAE	R^2
Linear Regression	19.0470	15.5454	0.0539
Random Forest	18.3106	14.2761	0.1256
XGBoost	18.0256	14.2637	0.1526

Although XGBoost emerged as the best-performing model with an R^2 of 0.1526, its predictive capacity remained limited. From an applied analytics perspective, an MAE of 14.26 points on a 0-to-100 popularity scale introduces relevant uncertainty: a track with an actual popularity of 50 could be estimated anywhere between approximately 36 and 64. These findings suggest that, although audio variables indicate statistically meaningful patterns, they explain only 15.26 percent of the variance in popularity. Therefore, the current dataset based on musical attributes is insufficient to predict a song's popularity with high precision, making it necessary to adopt a different approach to analyze the phenomenon.

4.2 Visualization of the Problem

The exploration of the dataset revealed a substantial concentration of records with zero popularity, a phenomenon technically known as zero-inflation. As illustrated in the popularity distribution histogram in Fig. 2, there is a high concentration at the value zero, which represents songs with no measurable popularity

in the dataset. Removing these records would be methodologically inappropriate, since it would ignore an important characteristic of the target distribution.

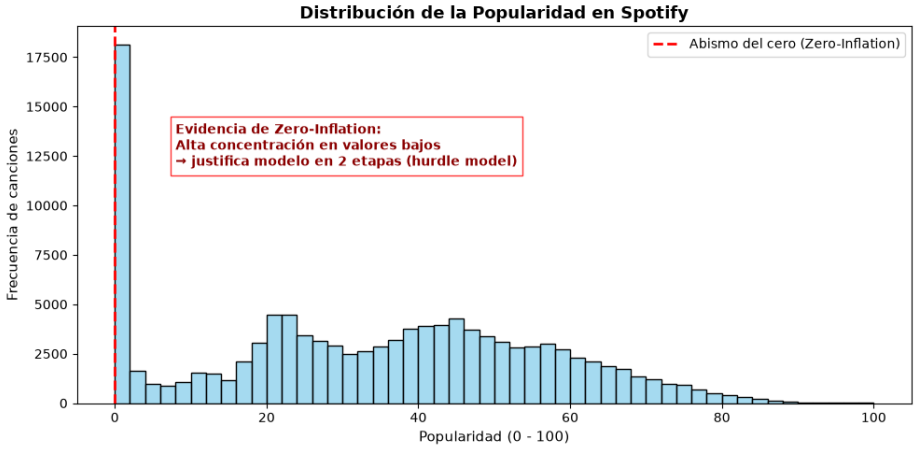


Fig. 2. Popularity distribution on Spotify.

This skewed distribution indicates that zero values and positive values may result from different underlying processes. While zero reflects the absence of measurable popularity in the dataset, positive values represent different degrees of platform visibility. This distinction transforms the nature of the problem from a direct numerical prediction task into a two-stage classification architecture known as the Two-Stage Hurdle Model. Consistent with previous work on zero-inflated outcomes [1,**?**], this approach first models the probability of overcoming the zero-popularity barrier before attempting to predict higher levels of popularity, beginning with the minimum relevance stage.

4.3 Minimum Relevance Classification

The first phase of the Hurdle architecture focused on identifying the factors that allow a song to move from zero popularity to positive popularity. This stage is essential to determine whether the models can identify whether musical characteristics contribute to a track's initial platform visibility. Table 2 compares the results of the models evaluated for this binary classification task.

The Random Forest model showed strong performance, reaching an F_1-score of 0.9631. Its recall value of 0.9904 is particularly relevant because it indicates that the model identified most songs with positive popularity. However, regularization adjustments should be explored in future iterations to reduce the risk of overfitting. Once the model's ability to recognize tracks with minimum relevance was identified, the next stage focused on predicting which of these tracks could become high-impact hits.

Table 2. Model performance in phase 1: zero versus positive popularity.

Model	Accuracy	Precision	Recall	F_1-Score
Logistic Regression	0.6005	0.9436	0.6083	0.7397
Decision Tree	0.5233	0.9551	0.5133	0.6677
Random Forest	0.9291	0.9372	0.9904	0.9631

4.4 Conditional Hit Prediction

After the initial relevance barrier is overcome, the objective becomes the prediction of high-impact hits. In this study, a hit was operationally defined as a song located in the 80th percentile of the distribution, equivalent to a popularity score of 54 according to the current data. This phase shows the difficulty of predicting high-popularity songs using only acoustic attributes, as shown in Table 3.

Table 3. Model performance in phase 2: hit or non-hit classification.

Model	Accuracy	Precision	Recall	F_1-Score
Decision Tree	0.7985	0.6292	0.0177	0.0344
Random Forest	0.8031	0.6316	0.0721	0.1294
XGBoost	0.8001	0.5968	0.0468	0.0868

In this second phase, the Random Forest model achieved the best relative performance, maintaining a global accuracy of 0.8031 and outperforming the other models. However, the recall values decreased substantially, reaching only 0.0721 for Random Forest. This low recall, accompanied by an F_1-score of only 0.1294 in the best case, confirms that although the models are effective at identifying many non-hit tracks, they are not able to detect high-impact hits with sufficient sensitivity using only the available audio features.

4.5 Evaluation of the Combined Hurdle Model

The performance of the final ensemble, the Combined RF Hurdle model, was evaluated after integrating the outputs of the two classification stages. The first stage estimated whether a track could overcome the zero-popularity barrier, while the second stage estimated whether the track could reach the top 20% of popularity. The final predictive capacity of the combined model is summarized in Table 4.

Figure 3 presents the visual evaluation of the combined Hurdle model. The ROC curve shows the model's ability to distinguish between hit and non-hit tracks across different thresholds, while the Precision-Recall curve provides a more informative perspective under class imbalance [2]. The probability distribution plot shows how the optimal threshold separates predicted positive and

Table 4. Final predictive capacity metrics.

Metric	Value
ROC-AUC	0.6865
Average Precision (AP)	0.3352
Optimal Threshold	0.3108

negative cases, and the confusion matrix summarizes the final classification results using this threshold.

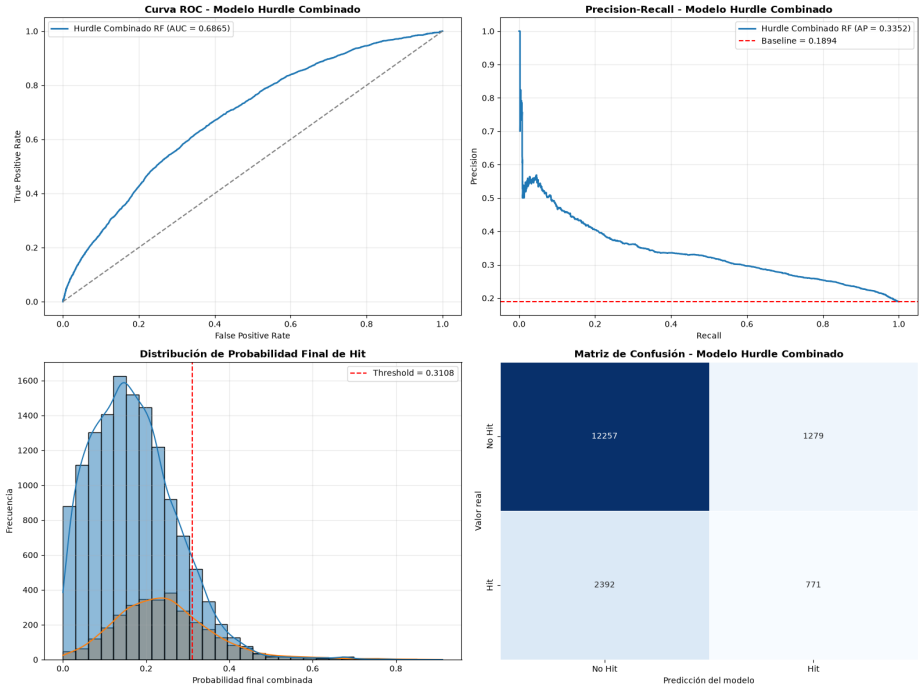


Fig. 3. Performance evaluation of the combined Hurdle model.

The ROC-AUC value of 0.6865 indicates that the model captures useful patterns for distinguishing songs with higher hit potential, although its performance is not strong enough to support fully automated decision-making. The Average Precision value of 0.3352 is particularly relevant because it evaluates the model under an imbalanced classification setting [2]. Since hit songs represent a smaller proportion of the dataset, this metric provides a more realistic view of the model's ability to prioritize promising tracks. Finally, the optimal threshold

of 0.3108 represents the decision point used to balance the detection of potential hits against the risk of incorrectly promoting non-hit tracks.

5 Discussion

5.1 Interpretation of the Models

From an applied music analytics perspective, the ROC-AUC value of 0.6865 indicates that the model captures patterns that may help identify songs with higher hit potential, but it does not include all the factors that determine whether a song becomes successful. In operational terms, this implies that the system may be useful for filtering large volumes of low-potential content and prioritizing songs that show initial commercial potential. However, given the number of external factors involved, the model should not be used as the sole criterion for real-world decision-making.

This is reinforced by the Average Precision value of 0.3352. When compared with the 18.94 percent baseline, which represents the probability of success by random selection in this dataset, the proposed pipeline achieves a 77 percent improvement in predictive efficiency. This suggests that the model extracts useful information from audio signals to identify popular songs, technically defined as tracks exceeding the popularity threshold of 54 points, corresponding to the 80th percentile.

The optimal threshold of 0.3108 represents a decision point at which the model balances the risk of missing a potential hit, that is, a false negative, against the risk of incorrectly classifying a non-hit track as a hit, that is, a false positive. The main limitation persists: acoustic variables are informative but not sufficient. Although features such as loudness and duration may contribute to popularity prediction, external market-related factors such as playlist placement, social media visibility, and marketing investment may strongly influence whether a song becomes successful.

5.2 Dataset Limitations

The dataset contains static musical characteristics that do not allow the model to determine dynamic factors such as the evolving popularity position of a song. In this case, several limitations can be identified, including the absence of a feature indicating the date on which the popularity value was collected, which would allow comparison with the current date. The dataset also lacks features capable of quantifying the previous popularity level of an artist, the degree of anticipation for a release, or a ratio that determines current relevance based on external factors. This limitation is especially relevant because Spotify defines track popularity as a platform-specific score calculated algorithmically and updated over time [10].

This static nature of the data generates a temporal bias that prevents the model from capturing the real life cycle of a musical product. Because the dataset

lacks longitudinal variables or time series, the model analyzes popularity as a fixed state rather than as a highly volatile variable dependent on context. Critical factors in the current market, such as the impact of viral marketing campaigns on platforms like TikTok, the inclusion of tracks in high-reach editorial playlists, or demand peaks derived from cultural events and concerts, remain outside the predictive scope of acoustic properties. Therefore, the current data architecture restricts the model to identifying commercially viable composition patterns, but it is insufficient to model the dynamic behavior of popularity in the digital music ecosystem.

6 Conclusions

This study showed that acoustic and musical features can provide useful information for identifying whether a song is likely to achieve a minimum level of relevance on Spotify. In the first stage of the proposed Hurdle strategy, the Random Forest model obtained strong classification performance, suggesting that measurable audio characteristics can help distinguish tracks with zero popularity from those with positive popularity. This result supports the idea that musical attributes are relevant for detecting initial market visibility.

However, the results also indicate that these features are not sufficient to accurately predict high-impact hits. Although the regression baseline and the second classification stage captured some patterns, their limited explanatory and predictive capacity shows that popularity is influenced by factors that are not represented in the static audio-feature dataset. Variables such as artist recognition, playlist placement, release timing, social media exposure, marketing investment, and cultural trends may play a decisive role in determining whether a song reaches the top popularity segment.

The main contribution of this study is the implementation of a reproducible two-stage machine learning pipeline that addresses Spotify popularity prediction under a zero-inflated target perspective. Instead of treating popularity only as a direct regression problem, the proposed approach separates the task into two more interpretable stages: first, identifying whether a track can overcome the zero-popularity barrier, and second, estimating whether it can reach the top 20% of popularity. This structure provides a more appropriate analytical framework for datasets where the target variable contains a high concentration of zero values.

Future work should incorporate contextual, temporal, and market-related variables to improve the prediction of high-impact hits. In addition, the use of cross-validation, repeated stratified validation, and hyperparameter optimization could strengthen the assessment of model stability and generalization. These extensions would allow the pipeline to move from an exploratory predictive framework toward a more robust decision-support tool for music analytics.

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